Ear Field Adaptive Noise Control for Snoring: An Real-time Experimental Approach

Lichuan Liu, *Senior Member, IEEE*, Karunakar Reddy Pottim, and Sen M. Kuo, *Senior Member, IEEE*

***Abstract*—In this paper, an active noise control (ANC) system is developed to provide an effective and non-intrusive solution for reducing loud snoring to provide a quiet environment for a snorer's bed partner. An adaptive least mean square (LMS) algorithm optimized for different kinds of snore signals is introduced and theoretically analyzed. Also, a residual noise masking approach is proposed to further reduce the effect of the snore noise without interfering with the LMS algorithm. Computer simulations followed by real-time experiments are conducted to demonstrate the feasibility of the snore ANC systems based on a pillow setup. For the optimum effect based on the characteristics of human hearing, the performance of the proposed approach is evaluated by using the multi-channel feedforward ANC systems based on the filtered-X least mean square (FXLMS) algorithm. Compared with a traditional headboard setup for snoring noise control, the proposed snore ANC systems optimized for ear field operation yield much higher noise reduction around the ears of the snorer's bed partner.**

***Index Terms*—Active noise control (ANC), audio integrated ANC, ear field, filtered-X least mean square (FXLMS), snore multi-channel.**

I. Introduction

SNORING is considered as a common sleep disorder, and about one third of adult population snores [1], [2]. In this paper, we use adaptive feedback control techniques to control snore noise problems, and propose an optimized least mean square (LMS) approach to cancel different kinds of snore signals [3], [4].

When a person inhales during sleep, air enters the mouth or nose and passes across the soft palate (the back of the roof of the mouth) on its way to the lungs. When the back of the mouth where the tongue and upper throat meet the soft palate and uvula collapses, the airway becomes narrow or even blocked. The narrowed or blocked passage disturbs the airflow, which causes the soft palate and uvula to vibrate and knock against the back of the throat, causing snoring [5], [6].

The average noise level of a snore signal is $60-80$ dB, which is much higher than 35dB, the night time noise level recommended by world health organization (WHO) [7]. Snoring can cause significant psychological and social damage to both snorers and their bed partners. Also, snoring can result in daytime drowsiness, irritability and reduced daytime alertness [8]. It is well known that snoring can cause sleep deprivation to the snorers when they suffer from sleep apnea. Moreover, snoring often cause sleep disturbances and other forms of stress for those around the snorers. Since snoring can be very loud, it can result in sleep deprivation of the snorer's bed partner [9]. The compromised quality and amount of sleep can directly affect the energy, concentration, and alertness of the snorer's bed partner, and consequently result in serious concerns, for example, loss of work productivity and occupational accidents. Research shows that reducing night-time sleep by as little as one and a half hours for just one night could result in a reduction of day-time alertness by as much as 32% [10], [11]. Snoring can also cause excessive bedroom noise exposure and may bring hearing problems to the snorers and their bed partner [11]. Furthermore, sleep loss could be a contributing factor for obesity. In addition, snoring could disturb the person's relationship with the bed partner, resulting in separate bedrooms, conflicts, etc [12].

Due to the lack of effective medical therapy to cure snoring, the existing medical techniques prescribed to control snoring would actually disturb the snorer's sleep [1], [2]. A non-invasive technique to reduce snoring noise for the bed partner without disturbing the snorer is necessary. Active noise control is an effective means of canceling low-frequency snoring noise based on the principle of destructive interference (i.e., the offending noise and the corresponding anti-noise superimpose destructively), resulting in the cancellation of each other in an acoustic domain [13]−[15]. The anti-noise is of the same amplitude and opposite in the phase of the offending noise, as shown in Fig. 1 [13]. Since most dominant frequency contents of snores are below 2 KHz, an ANC system can effectively reduce the snoring noise for the snorer's bed partner.

However, current snore ANC approaches are based on a general snore noise model with a headboard setup [16], [17]. Considering that each snorer's noise has unique acoustic characteristics, a real time adaptive algorithm for snore noise cancellation is necessary. Also, for the headboard setup, the quiet zone centered at the error microphones is away from the ears of the snorer's bed partner, resulting in less effective noise reduction around the ears of the snorer's bed partner. Moreover, a headboard is heavy, bulky and not portable. To solve these problems, we propose a novel approach with real time adaptive capability for different kinds of snore signals, where the error microphones will be placed inside the pillow

Manuscript received June 26, 2018; accepted September 14, 2018. Recommended by Associate Editor Mingcong Deng. *(Corresponding author: Lichuan Liu.)*

Citation: L. C. Liu, K. R. Pottim, and S. M. Kuo, "Ear field adaptive noise control for snoring: an real-time experimental approach," *IEEE/CAA J. Autom. Sinica*, vol. 6, no. 1, pp. 158−166, Jan. 2019.

L. C. Liu and K. Reddy Pottim are with the Electrical Engineering Department, Northern Illinois University, DeKalb IL 60115 USA (email: liu@niu.edu; z15 78026@niu.edu).

S. Kuo is with the Department of Electrical Engineering, Chung Yuan Christian University, Taipei 32023, China (email: smkuo@cycu.edu.tw).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/JAS.2019.1911339

Authorized licensed use limited to: Utah State University. Downloaded on February 03,2026 at 03:43:34 UTC from IEEE Xplore. Restrictions apply.

of the snorer's bed partner, which ensures that the error microphones are close to the ears of the snorer's bed partner to construct an effective ear field quiet zone [18]. Furthermore, the secondary loudspeakers are also placed inside the pillow. In this way, the ANC system based in the pillow requires lower volume of anti-noise. This reduces acoustic feedback to the reference microphone, as compared to the headboard setup, and this improves the accuracy and performance. Therefore, users can take this portable snore ANC system with him/her. The proposed system can also reduce other kinds of unwanted noise, such as, air condition, heating, and traffic noise.

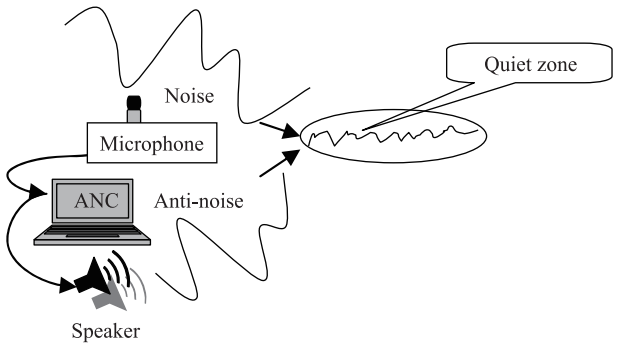


Fig. 1. Principle of ANC system.

The snore ANC system provides an effective solution to reduce loud snores for the snorer's bed partner. However, the ANC system does not result in complete cancellation of snoring noise. The residual noise could be still audible in a quiet environment. Sound masking is a method that mixes an artificial or natural sound with the anti-noise to mask the unwanted residual snore (sound masking). Thus, we can minimize the effect of distracting residual snores using sound masking [19]. At the same time, as the residual noise signals are used to update the adaptive filters, the proposed masking approach should not interfere with this function.

The rest of paper is organized as follows. Section II proposes the optimized adaptive LMS algorithm and also the multi-channel FXLMS algorithm for active control. Section III introduces an audio integrated ANC system to mask the residual noise. Section IV presents the computer simulation results for both the headboard and pillow setup. Section V presents real-time experiments using Knowles Electronics Manikin for Acoustic Research (KEMAR) to verify the performance of the proposed ANC systems for practical applications. Section VI concludes the paper.

II. Multi-channel Snore ANC System

A. Adaptive ANC Algorithms

This section introduces the ANC algorithms proposed for snoring noise cancellation. Snore signal is a time-varying and nonstationary random signal, therefore, a wideband feedforward ANC system is used for snore signal cancellation [13], [15]. On the other hand, the quiet zone generated by the proposed ANC system is centered at the error microphone, cancellation performance decreases as the distance from the error microphone increases [13]. So, we propose the use of multi-channel feedforward ANC system to provide multiple (two) quiet zones around the bed partner's two ears.

The block diagram of a multi-channel ($J \times M \times K$) feedforward ANC system is shown in Fig. 2. We assume that there are J reference microphones which are used to pick up the primary noise signals (snore signals in this paper), K secondary sources which are the anti-noise speakers used for generating anti-noise signals, and M error microphones which pick up residual noise signals. The residual noise signals are used to update the adaptive filters. The reference signals from J reference microphones can be expressed as

$$x(n) = [x_1^T(n) \quad \dots \quad x_J^T(n-L+1)]^T$$

where $x_j^T(n)$ is the jth reference signal of length L. The secondary sources have K channels, $y(n) = [y_1(n) \ \dots \ y_K(n)]^T$, where $y_k(n)$ is the signal of kth output channel at time n. The error signals have M channels, $e(n) = [e_1(n) \ \dots \ e_M(n)]^T$, where $e_m(n)$ is the error signal from the mth error channel at the time n.

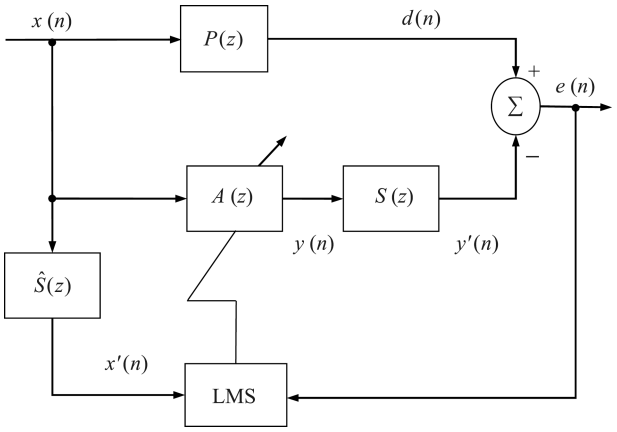


Fig. 2. Block diagram of multi-channel ANC system using FXLMS algorithm.

We model the unknown acoustic paths from the reference microphones to the error microphones as the primary paths, $P(z)$. The secondary paths $S(z)$ include the transfer functions of the digital-to-analog converter, reconstruction filter, power amplifier, loudspeaker, acoustic path from the loudspeaker to the error microphone, error microphone, preamplifier, anti-aliasing filter, and analog-to-digital converter [13]. $\hat{S}(z)$ is the estimate of the secondary path, $S(z)$. The adaptive filter, $A(z)$, generates the anti-noise, $y(n)$, which is the output through the secondary loudspeaker. The actual anti-noise signals $y'(n)$ at the error microphones $y(n)$ pass through the secondary paths. Both the primary noise $d(n)$ and the canceling noise $y'(n)$ are vectors with M elements at the locations of M error microphones [13].

The error signals $e(n)$ are residual noise signals picked up by the error microphones. The error signal is used to update the coefficients of the adaptive filter $A(z)$, based on the filtered-X least mean square (FXLMS) algorithm [11].

The primary noise to be cancelled/reduced have M channels which can be expressed by

$$d(n) = \begin{bmatrix} d_1(n) & d_2(n) & \cdots & d_M(n) \end{bmatrix}^T$$

Authorized licensed use limited to: Utah State University. Downloaded on February 03,2026 at 03:43:34 UTC from IEEE Xplore. Restrictions apply.

$$= \underbrace{\begin{bmatrix} p_{11}(n) & p_{12}(n) & \cdots & p_{1J}(n) \\ p_{21}(n) & p_{22}(n) & \cdots & p_{2J}(n) \\ \vdots & \vdots & \ddots & \vdots \\ p_{M1}(n) & p_{M1}(n) & \vdots & p_{MJ}(n) \end{bmatrix}}_{P(n)} * \underbrace{\begin{bmatrix} x_1^T(n) \\ x_2^T(n) \\ \vdots \\ x_J^T(n) \end{bmatrix}^T}_{x(n)}. \tag{1}$$

The primary paths impulse responses can be expressed by a matrix as $P(n)$ with $p_{mj}(n)$ the impulse response function from the jth reference microphone to the mth error microphone. The matrix of secondary path impulse response functions is

$$S(n) = \begin{bmatrix} s_{11}(n) & s_{12}(n) & \cdots & s_{1K}(n) \\ s_{21}(n) & s_{22}(n) & \cdots & s_{2K}(n) \\ \vdots & \vdots & \ddots & \vdots \\ s_{M1}(n) & s_{M2}(n) & \cdots & s_{MK}(n) \end{bmatrix}$$

where $s_{mk}(n)$ is the impulse response function from the kth secondary source to the mth error microphone. An estimate of $S(n)$, denoted as $\hat{S}(n)$, can be similarly defined.

Matrix $A(n)$ consists of feedforward adaptive finite impulse response (FIR) filter impulse response functions, which has J inputs, K outputs, and filter order L, can be denoted as

$$A[n] = [A_1^T[n] \quad \dots \quad A_J^T[n]]^T$$

where $A_k[n] = [A_{k,1}^T[n] \ \dots \ A_{k,J}^T[n]]^T, k = 1, 2, \dots, K$ is the weight vector of the kth feedforward FIR adaptive filter with J input signals $A_{k,j}[n] = [a_{k,j,1}[n] \ \dots \ a_{k,j,L}[n]]^T$, which is the feedforward FIR weight vector form the jth input to kth output.

The secondary sources are driven by the feedforward filters' outputs. That is

$$y_k(n) = \sum_{j=1}^{J} x_j^T(n) A_{k,j}(n) = x^T(n) A_k(n). \tag{2}$$

The error signal vector measured by M microphones is

$$\begin{aligned} e(n) &= d(n) - y'(n) \\ &= d(n) - S(n) * [X^T(n) A(n)] \end{aligned} \tag{3}$$

where $d(n)$ is the primary noise vector and $y'(n)$ is the anti-noise pass at the error microphones.

The filter coefficients are iteratively updated to minimize a defined criterion. We use the sum of the mean square errors as the cost function defined as

$$\xi(n) = \sum_{m=1}^{M} E\left\{e_m^2(n)\right\} = e^T(n) e(n). \tag{4}$$

The least mean square (LMS) adaptive algorithm uses a steepest descent approach to adjust the coefficients of the feedfoward adaptive FIR filters in order to minimize $\xi(n)$ as follows [9]:

$$A(n+1) = A(n) - \mu_a X(n) e(n) \tag{5}$$

where μ_a is the step size for feedforward ANC systems. In real-time experiments, we use different values to improve the convergence speed.

The filtered reference signals can be calculated as follows:

$$\begin{aligned} X'(n) &= \left[S(n) * X^T(n)\right]^T \\ &= \left[\begin{bmatrix} \hat{s}_{11}(n) & \hat{s}_{12}(n) & \cdots & \hat{s}_{1K}(n) \\ \hat{s}_{21}(n) & \hat{s}_{22}(n) & \cdots & \hat{s}_{2K}(n) \\ \vdots & \vdots & \ddots & \vdots \\ \hat{s}_{M1}(n) & \hat{s}_{M2}(n) & \cdots & \hat{s}_{MK}(n) \end{bmatrix} \right. \\ &\quad \left. * \begin{bmatrix} x(n) & 0 & \cdots & 0 \\ 0 & x(n) & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \cdots & x(n) \end{bmatrix}^T \right]^T \\ &= \begin{bmatrix} x'_{11}(n) & x'_{12}(n) & \cdots & x'_{1M}(n) \\ x'_{21}(n) & x'_{22}(n) & \cdots & x'_{2M}(n) \\ \vdots & \vdots & \ddots & \vdots \\ x'_{K1}(n) & x'_{K2}(n) & \cdots & x'_{KM}(n) \end{bmatrix} \end{aligned}$$

and we could have

$$\begin{aligned} x'_{km}(n) &= s_{mk(n)} * x(n) \\ &= \left[s_{mk(n)} * x_1^T(n) s_{mk(n)} * x_2^T(n) \cdots s_{mk(n)} * x_J^T(n)\right] \\ &= \left[x'^T_{km1}(n) x'^T_{km2}(n) \cdots x'^T_{kmJ}(n)\right]. \end{aligned}$$

The updated adaptive filter's coefficients are [14]

$$A_k(n+1) = A_k(n) - \mu \sum_{m=1}^{M} x'_{km}(n) e_m(n) \tag{6}$$

it can be further expanded as

$$A_{k,j}(n+1) = A_{k,j}(n) - \mu \sum_{m=1}^{M} x'_{km}(n) e_m(n). \tag{7}$$

Therefore, with the ANC structure and also the coefficients above, we can generate quiet zones around the error microphones.

B. Modified Adaptive LMS Algorithm (ALMS)

For stochastic input, the LMS algorithm uses deterministic step size and filter length in a recursive computation of the Wiener filter. Therefore, a significant feature of the LMS algorithm is its simplicity, which makes real time processing possible in the proposed approach. However, since snore signals are time-varying and non-stationary signals, traditional LMS algorithms cannot reduce snore noise effectively. Moreover, since each snorer's snore signals have their unique time-frequency characteristics, so it is essential to design an adaptive LMS algorithm for the best cancellation performance for different snore signals.

First, adaptive step sizes are employed to update the filter. Compared with LMS, ALMS balances fast convergence and mis-adjustment well. It limits the influence from the input noise which can lead to divergence and improves the properties for the system with moderate computational complexity.

An adaptive step size $\mu(n)$ for snore noise signal can be expressed as

Authorized licensed use limited to: Utah State University. Downloaded on February 03,2026 at 03:43:34 UTC from IEEE Xplore. Restrictions apply.

$$\mu(n) = \frac{1}{\frac{1}{N}\sum_{m=0}^{N-1}[w(m)x(n-m)]^2} \tag{8}$$

where $w(m)$ are the coefficients of a window function, m stands for window index and n stands for index of sample. The length of the window function N can be calculated as

$$N = \frac{T_f}{T_s} = \frac{F_s}{F_f} \tag{9}$$

where F_f is the fundamental frequency of snore signals which is determined by the anatomy structure of the snorer and F_s is the sampling frequency.

For an optimum LMS algorithm for snore signal cancellation, the filter length is also adaptive to the input snore signals:

$$L = \alpha N \tag{10}$$

where α is a scale with $1 < \alpha < 5$, which will be selected in the training stage based on the input snore signal.

III. Audio Integration in ANC Systems

In this section, audio-integrated ANC algorithms introduce desired sound to mask the residual noise. The masking audio is chosen to provide a peaceful and soothing environment for the snorer's bed partner. Natural sounds like rain, a water stream, and a waterfall are good cases of maskers [19]. Most of the natural sounds have a flat frequency spectrum. This characteristic implies the absence of dominating frequency components. This also helps in the relaxation of human brain and does not disturb the sleeper [19].

In audio integration, the masking sound is added to the anti-noise and the mixed signal is used to drive the secondary loudspeaker. The masking sound with the residual noise is picked up by the error microphones. Without special treatment, the masking sound will act as interference that degrades the performance of the FXLMS algorithm. However, only the actual error signal (residual noise) is supposed to be used for the adaptation of the ANC algorithm. In addition, the masking sound should not be canceled by the ANC system. The algorithm of the audio-integrated ANC system is shown in Fig. 3.

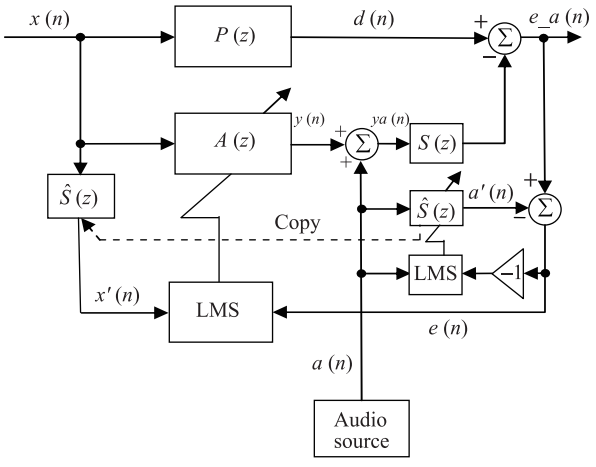


Fig. 3. Block diagram of the audio-integrated ANC system.

In the audio-integrated algorithm, the masking signal $a(n)$ is added to the anti-noise $y(n)$. The combined signal $ya(n)$ is used to drive the secondary loud speaker. The error signal $e_a(n)$ picked up by the error microphone is a combination of the "true" error signal $e(n)$ and the audio component $a(n)$ filtered through the secondary path $S(z)$. To separate the masking audio from the "true" error signal, an estimate of the masking audio, $a'(n)$, is generated by filtering $a(n)$ with $\hat{S}(z)$. This estimate is subtracted from the measured error signal $e_a(n)(n)$ to obtain the "true" error (residual noise) $e(n)$ for updating the LMS algorithm [11].

Error signals in the z-domain could be expressed as [20]

$$E_A(z) = D(z) - S(z)[Y(z) + A(z)] \tag{11}$$

and

$$\begin{aligned} E(z) &= E_A(z) + \hat{S}(z)A(z) \\ &= D(z) - S(z)Y(z) + [\hat{S}(z) - S(z)]A(z). \end{aligned} \tag{12}$$

Assuming that $S(z) \approx \hat{S}(z)$ and that the audio signal is uncorrelated with the primary noise, we have

$$E(z) \approx D(z) - S(z)Y(z). \tag{13}$$

Equation (10) can be expressed in the time domain as

$$e(n) \approx d(n) - s(n) * y(n). \tag{14}$$

This is the "true" error signal $e(n)$ used to update the LMS algorithm. The audio integrated ANC system uses the same existing hardware utilized by ANC system, so it will not increase the hardware cost.

IV. Experiment Results

A. Experimental Setup

This section introduces the ANC experimental setup for an ear field environment, such as on a pillow. The details about the headboard setup, which we could use as a performance reference, can be found in [16], [17]. The experimental setup of the multi-channel feedforward snore ANC system using the $1 \times 2 \times 2$ (1 reference, 2 secondary speakers and 2 error microphones) FXLMS algorithm is shown in Fig. 4. In this setup, two error microphones and two secondary loudspeakers are placed inside the pillow as sketched in Fig. 5, where the loudspeakers are located at the back corners of pillow and microphones are placed near the top of the pillow which would be under the ears of the bed partner. The dimension of the pillow used for the experiment is $19.5'' \times 15.5'' \times 5''$. The error microphones are located at a depth of about $1''$ inside the pillow, under the ears of the KEMAR, which represents the bed partner. The secondary loudspeakers are located at the back corners of the pillow, about $8''$ away from the error microphones. The recorded snore is played through the primary loudspeaker which represents the snorer. KEMAR has two microphones in its ears and these microphones are used to measure the snore received at the ears of the snorer's bed partner. In Fig. 4, we show the head board setup.

The distance between the left (or right) error microphone and the corresponding ear of the KEMAR for the pillow setup,

Authorized licensed use limited to: Utah State University. Downloaded on February 03,2026 at 03:43:34 UTC from IEEE Xplore. Restrictions apply.

TABLE I
NOISE REDUCTION AT DIFFERENT TONAL FREQUENCIES FOR THE MULTI-CHANNEL FEEDFORWARD ANC SYSTEM (H MEANS HEADBOARD SETUP AND P MEANS PILLOW SETUP)

Frequency (Hz)	Noise reduction (dB)							
	Right error microphone		Right ear of the KEMAR		Left error microphone		Left ear of the KEMAR	
Setup	H	P	H	P	H	P	H	P
50	10	10	7	12	9	12	7	14
100	22	31	18	27	21	29	19	39
150	34	44	30	42	32	46	29	44
200	42	52	36	54	42	48	36	46
250	51	55	45	52	47	51	41	49
300	45	49	37	47	46	48	38	47
350	31	37	27	38	34	34	31	33
400	28	34	22	32	21	28	18	30
450	15	18	13	16	13	16	11	14
500	10	12	7	7	9	12	7	13
600	6	7	2	9	3	6	2	3
Average noise reduction	27	32	22	31	25	30	21	30

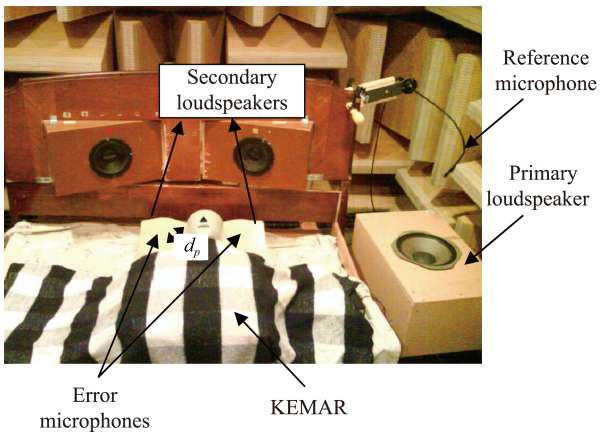


Fig. 4. Experimental setup for active snore control system.

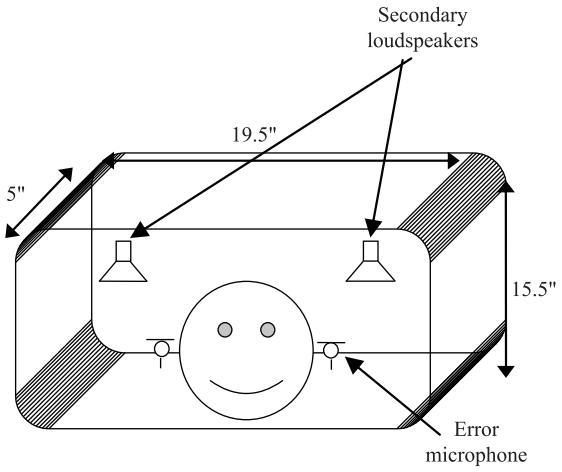


Fig. 5. The pillow setup, top view.

represented by d_P, is about 4 inches, which is much shorter than the headboard setup. As the quiet zone produced by the ANC is centered at the error microphone and the noise reduction effect decreases as we move away from the center of the quiet zone, the noise reduction at the ears of the KEMAR for the pillow setup will be higher than the headboard setup.

The primary-path transfer functions and the secondary-path transfer functions are obtained for computer simulations. These transfer functions are measured inside an acoustic chamber using an HP dynamic signal analyzer for both setups. A random noise generated by the dynamic signals analyzer is played through the primary loudspeaker. The primary-path transfer functions, $P_1(z)$ (from the primary loudspeaker to the right error microphone) and $P_2(z)$ (from the primary loudspeaker to the left error microphone) are measured by the analyzer. Another set of primary-path transfer functions from the primary loudspeaker to the two ears of the KEMAR are also estimated. The sampling frequency for these measurements is 2 KHz. To acquire the secondary-path transfer functions, the same random noise generated by the analyzer is played through the secondary loudspeakers. The secondary-path transfer functions from the two secondary loudspeakers to the error microphones and the ears of the KEMAR are measured by the analyzer. The transfer functions recorded in the spectrum analyzer can be plotted by using MATLAB.

B. Computer Simulation

The ANC algorithms discussed in Section II are evaluated by computer simulations using both tonal noise and recorded snore noise as the primary noise. The ANC system is first tested for tonal noise at different frequencies, and the simulation results are discussed. In computer simulation, the filter length for secondary path is 64 with step size 0.1, the filter length for adaptive filter is 128, and the step size is 0.00009.

Table I lists the noise cancellation achieved by the multi-channel feed-forward ANC system based on the headboard and pillow setups. An average reduction of 27 dB is achieved at the right error microphone and 25 dB is achieved at the left error microphone for the headboard setup. For the pillow setup, an average reduction of 32 dB is achieved at the right error microphone and 30 dB is achieved at the left error microphone.

Authorized licensed use limited to: Utah State University. Downloaded on February 03,2026 at 03:43:34 UTC from IEEE Xplore. Restrictions apply.

The noise reduction achieved at the right and left ears of the KEMAR are 31 dB and 30 dB for the pillow setup; while 22 dB and 21 dB are achieved using the headboard. The performance of the multi-channel ANC system for the pillow setup is better as compared to the headboard setup. Also, the noise reduction achieved at the error microphones is similar to the ears of the KEMAR because the ears of the KEMAR are located closer to the error microphones.

The computer simulations of the multi-channel feedforward ANC system using the $1 \times 2 \times 2$ FXLMS algorithm are conducted for the case where the primary noise consists of snore noise. The ANC system is tested for both the headboard and pillow setups.

The spectra of the error signals before and after ANC at the right error microphone and the right ear of the KEMAR for the headboard setup are shown in Fig. 6. Average noise reduction of 16 dB and 13 dB are achieved at the right error microphone and right ear of the KEMAR, respectively.

In Fig. 7, the spectra of signals before and after ANC measured at the right error microphone and the right ear of KEMAR based on the pillow setup are presented. An average noise reduction of 21 dB is achieved at the right error microphone and 19 dB is achieved at the right ear of KEMAR.

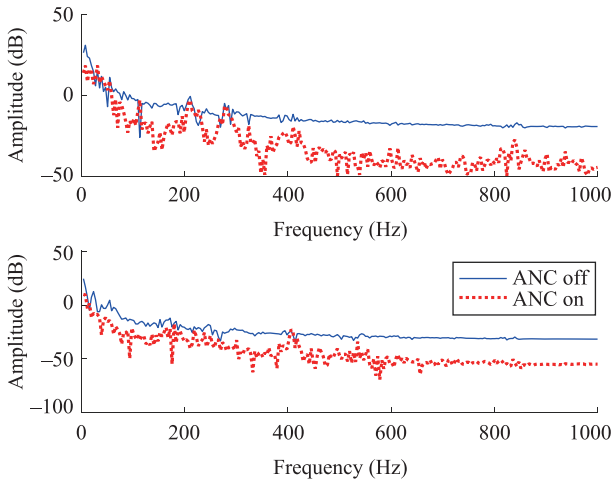


Fig. 6. Spectra of signals at (upper) right error microphone and (lower) right ear of KEMAR before and after ANC for the headboard setup.

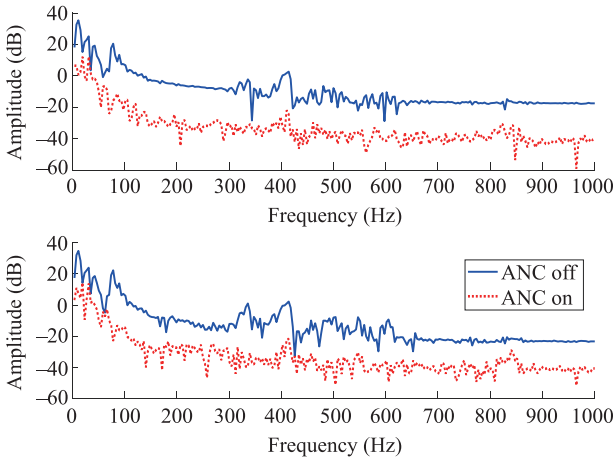


Fig. 7. Spectra of signals at (upper) right error microphone and (lower) right ear of KEMAR before and after ANC for the pillow setup.

The noise reduction achieved by the proposed adaptive algorithm with the ear field (pillow) setup is much higher than the headboard setup, and the noise reduction at the error microphones is similar to the ears of the KEMAR based on the pillow setup. This is because the distance between the ears of the KEMAR and the center of the quiet zone at the error microphones is shorter. We can tell in Table II that the adaptive LMS has much better performance than the traditional LMS.

TABLE II
ATTENUATION OF DIFFERENT LMS METHODS IN dB

Algorithm	Attenuation (L)	Attenuation (R)
ALMS	19.7	20.9
LMS	16.1	12.1

In Fig. 8, we can find that the noise cancellation for ALMS algorithm is 19 dB at left side and 20 dB at right side.

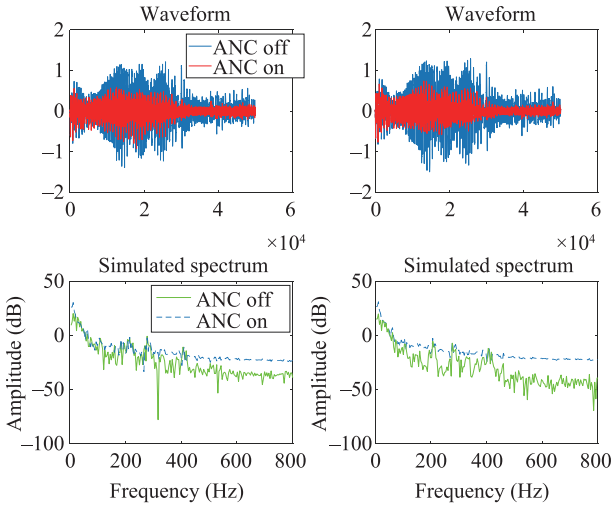


Fig. 8. Snore cancellation performance for ALMS algorithm, (upper left) time domain waveform at left error microphone (upper right) waveform at the right error microphone, (lower left) spectrum of signals at left error microphone, (lower right) spectrum of signals at right error microphone.

In Fig. 9, we can find that the noise cancellation for LMS algorithm is 16 dB at left side and 12 dB at right side.

V. REAL TIME EXPERIMENT

In this section, the experiment results of the multi-channel feedforward ANC system using the $1\times2\times2$ FXLMS algorithm are presented. The system is tested using two experimental setups, one is based on the traditional headboard setup and the other is based on the proposed adaptive LSM system with the pillow set up. The real-time experiment is conducted on a DSP platform we established at the anechoic chamber at Northern Illinois University in order to minimize the reflection of sound waves and other acoustic interferences, as shown in Fig. 10.

In the real-time experiment, the filter length for secondary path is 64 with step size 0.02, the filter length for adaptive filter is 128, and the step size for update $W(z)$ is 0.001. The multi-channel ANC system is first tested using sinusoidal primary noise at frequencies varying from 50 Hz to 600 Hz.

For the headboard setup, an average noise reduction of

Authorized licensed use limited to: Utah State University. Downloaded on February 03,2026 at 03:43:34 UTC from IEEE Xplore. Restrictions apply.

TABLE III
NOISE REDUCTION AT DIFFERENT TONAL FREQUENCIES FOR THE MULTI-CHANNEL FEEDFORWARD ANC SYSTEM (H MEANS HEADBOARD SETUP AND P MEANS PILLOW SETUP)

Frequency (Hz)	Noise reduction (dB)							
	Right error MIC		Right Ear		Left error MIC		Left Ear	
Setup	H	P	H	P	H	P	H	P
50	9	9	5	10	9	11	6	11
100	21	33	16	28	17	24	15	29
150	24	28	21	34	21	29	18	27
200	31	36	26	39	30	34	27	28
250	35	31	30	32	31	32	28	26
300	41	45	35	48	39	41	35	44
350	25	28	22	26	22	22	20	28
400	17	21	13	17	18	28	15	22
450	13	15	11	18	11	9	10	12
500	10	11	9	9	10	12	7	10
600	9	9	5	5	7	4	6	8
Average noise reduction	24	24	18	24	22	22	17	22

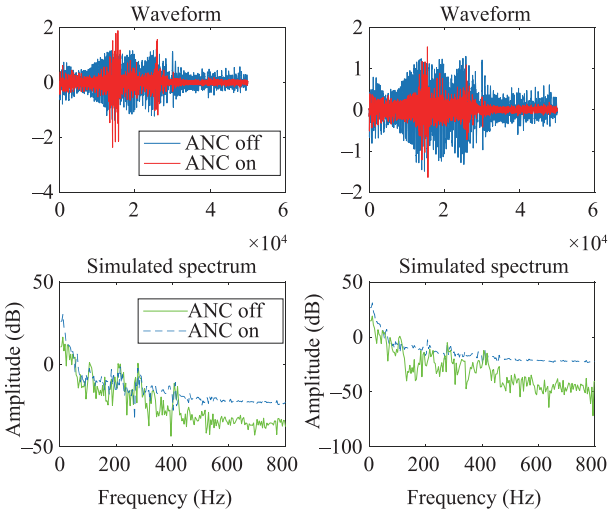


Fig. 9. Snore cancellation performance for LMS algorithm, (upper left) time domain waveform at left error microphone (upper right) waveform at the right error microphone, (lower left) spectrum of signals at left error microphone, (lower right) spectrum of signals at right error microphone.

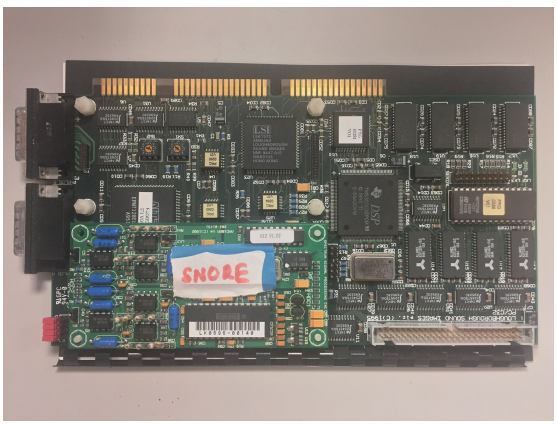


Fig. 10. The DSP platform for real-time snore cancellation experiment.

24 dB is achieved at the right error microphone and 18 dB is achieved at the right ear of the KEMAR. An average noise reduction of 22 dB is achieved at the left error microphone and 17 dB is achieved at the left ear of the KEMAR. As shown in Table III, an average noise reduction of 24 dB is achieved at both the right error microphone and the right ear of the KEMAR. An average noise reduction of 22 dB is achieved at the left error microphone and the left ear of the KEMAR for the pillow setup. It is noted that the noise cancellation at the error microphones is much higher than the ears of the KEMAR for the headboard setup; while the noise cancellation is similar at the error microphones and the ears of the KEMAR for the pillow setup.

The multi-channel feedforward ANC system is further tested using the recorded snore for both setups. The spectra of the error signals before and after ANC at the right error microphone and the right ear of the KEMAR for the headboard setup are shown in Fig. 11. Average noise reduction of 16 dB

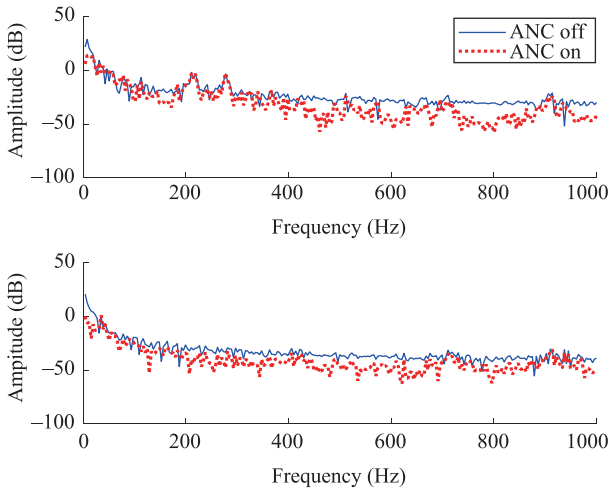


Fig. 11. Spectra of signals at (upper) right error microphone and (lower) right ear of KEMAR before and after ANC for the headboard setup.

Authorized licensed use limited to: Utah State University. Downloaded on February 03,2026 at 03:43:34 UTC from IEEE Xplore. Restrictions apply.

and 13 dB are achieved at the right error microphone and right ear of the KEMAR, respectively.

The spectra of the error signals before and after ANC at the left error microphone and the left ear of the KEMAR for the pillow setup are shown in Fig. 12. An average noise reduction of 18 dB and 17 dB is achieved at the left error microphone and left ear of the KEMAR, respectively.

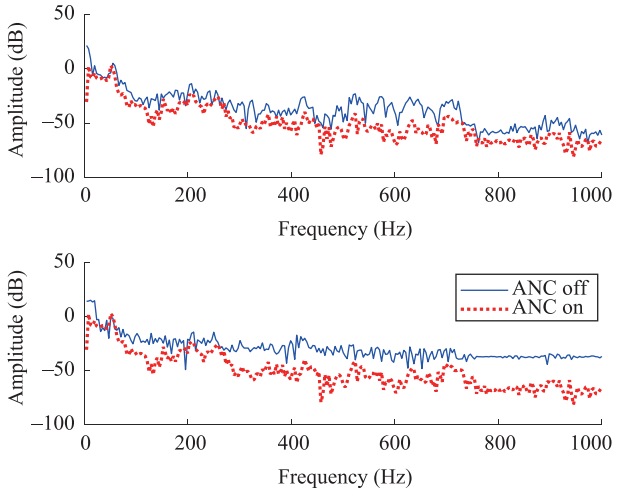


Fig. 12. Spectra of signals at the (upper) left error microphone and (lower) left ear of KEMAR before and after ANC for the pillow setup.

The Audio-integrated ANC has been also tested by real-time experiment. The masking audio used in testing the audio-integration is the sound of a river stream. The time-domain waveform of the masking signal is shown in Fig. 13. The ANC system does not affect noise cancellation performance, and the audio signal can mask snore sound effectively. Based on the results shown in this section, it is observed that the proposed ANC system with the pillow setup outperforms traditional algorithms with the headboard setup.

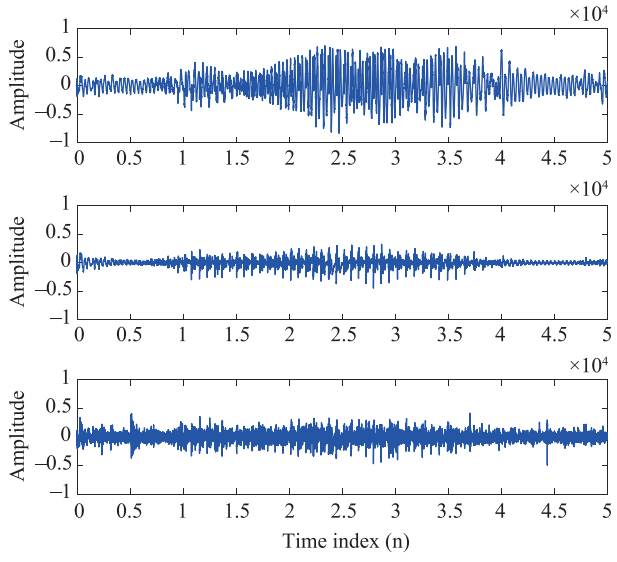


Fig. 13. Audio integrated ANC systems (upper) the snore noise wave form, (middle) the wave form of the audio signal and (lower) the waveform of the masked residual noise

VI. Conclusion

In this paper, an adaptive real time multi-channel feedforward ANC algorithms for snore noise cancellation with ear field (pillow) setup was proposed. An algorithm to introduce masking sound for a better user experience without interfering with the noise cancellation function has also been proposed. The proposed algorithms have been given in detailed mathematical models and verified by computer simulations and testbed experiments. The proposed system with an ear field setup outperforms the original headboard setup by an average of 4 dB at the KEMAR's ears for real recorded snore noise. Experiments also showed that the multi-channel feedforward ANC system based on the pillow setup successfully canceled the snore noises and performed consistently better than the headboard setup in real time operation. The audio integrated ANC system has been proven to mask the residual noise in experiments.

References

[1] K. Ramar, L. C. Dort, S. G. Katz, C. J. Lettieri, C. G. Harrod, S. M. Thomas, and R. D. Chervin, "Clinical practice guideline for the treatment of obstructive sleep apnea and snoring with oral appliance therapy: an update for 2015, an american academy of sleep medicine and american academy of dental sleep medicine clinical practice guideline," *J. Clinical Sleep Med.*, vol. 1, no. 7, pp. 773−827, Jul. 2015.

[2] K. A. Franklin and E. Lindberg, "Obstructive sleep apnea is a common disorder in the population: a review on the epidemiology of sleep apnea," *J. Thoracic Disease*, vol. 7, no. 8, pp. 1311−1322, Aug. 2015.

[3] N. Bu, W. Chen, L. Jin, and Y. Zhao, "Robust control for uncertain nonlinear feedback system using operator-based right coprime factorization," *IEEE/CAA J. Autom Sinica*, pp. 1−6, 2018, doi: 10.1109/JAS.2017.7510895.

[4] C. Ji, J. Wang, L. Cao, and Q. Jin, "Parameters tuning of model free adaptive control based on minimum entropy," *IEEE/CAA J. Autom. Sinica*, vol. 1, No. 4, pp. 361−371, Oct. 2014.

[5] K. Qian, C. Janott, V. Pandit, Z. Zhang, C Heiser, W. Hohenhorst, M. Herzog, W. Hemmert, and B. Schuller, "Classification of the excitation location of snore Sounds in the upper airway by acoustic multifeature Analysis," *IEEE Trans. on Biomedical Engineering*, vol. 64, no. 8, pp. 1731−1741, Oct. 2017.

[6] L. Huang, "Mechanical modeling of palatal snoring," *J. Acoust. Soc. Am.*, vol. 97, no. 6, pp. 3642−3648, Jun. 1995.

[7] B. Berglund, T. Lindvall, and D. H. Schwela, "Guidelines for a community noise," *WHO Document*, Apr., 1999.

[8] E. Lindberg, N. Carter, T. Gislason, and C. Janson, "Role of snoring and daytime sleepiness in occupational accidents," *Am. J. Respir. Crit. Care Med.,* vol. 164, pp. 2031−2035, Dec. 2001.

[9] S. Adam, L. Geiss, L. Peter, G. Niklewski, and K Richter, "Two in a bed: the influence of couple sleeping and chronotypes on relationship and sleep: an overview," *Chronobiol Int.,* vol. 33, no. 10, pp. 1464−1472, Nov. 2016.

[10] A. J. Krause, E. B. Simon, B. A. Mander, S. M. Greer, J. M. Saletin, A. N. Goldstein-Piekarski, and M. P. Walker, "The sleep-deprived human brain," *Nature Reviews Neuroscience*, vol. 18, pp. 404−418, May 2017.

[11] J. C. Lo, J. L. Ong, R. L. F. Leong, J. J. Gooley, and M. W. L. Chee, "Cognitive performance, sleepiness, and mood in partially sleep deprived adolescents: the need for sleep study," *Sleep*, vol. 39, no. 3, pp. 687−698, Mar. 2016.

[12] M. Breus, *Beauty Sleep: Look Younger, Lose Weight, and Feel Great Through Better Sleep,* Plume, New York, NY, USA, 2007.

[13] S. M. Kuo and D. R. Morgan, "Active noise control: a tutorial review," *Proc. the IEEE*, vol. 87, pp. 943−973, Jun. 1999.

Authorized licensed use limited to: Utah State University. Downloaded on February 03,2026 at 03:43:34 UTC from IEEE Xplore. Restrictions apply.

[14] K.-W. Liang and J.-S. Hu, "An open-loop pole-zero placement method for active noise control headphones," *IEEE Trans. on Control Systems Technology*, vol. 25, no. 4, pp. 1278−1283, 2017.

[15] S. Li, M. C. Zhou, and X. Luo, "Modified primal-dual neural networks for motion control of redundant manipulators with dynamic rejection of harmonic noises," *IEEE Trans. Neural Networks and Learning Systems*, vol. pp, no. 99, pp. 1−11, Dec. 2017.

[16] S. M. Kuo and S. R. Chakravarthy, "Application of active noise control for reducing snore," in *Proc. Int. Conf. on Acoustics, Speech, and Signal Processing*, Toulouse, France, 2006, pp. 305−308.

[17] S. M. Kuo and R. Gireddy, "Real-time experiment of snore active noise control," in *Proc. IEEE Conf. on Contr. Application*, Singapore, Singapore, 2007, pp. 1342−1346.

[18] S. Kuo, C. Chang, K. Pottim, and L. Liu "Active snore control system integrated with apnea detector", in *InterNoise*, Melbourne, Australia, 2014, pp. 1−7.

[19] T. Renza, "Auditory distraction by speech: sound masking with speech-shaped stationary noise outperforms −5 dB per octave shaped noise," *J. Acoustical Society of America,* vol. 143, no. 3, EL212, 2018.

[20] C. Chang, S. M. Kuo, and C. W. Huang, "Secondary path modeling for narrowband active noise control systems," *Applied Acoustics*, vol. 131, pp. 154−164, Feb. 2018.



Lichuan Liu (M'01−SM'11) received her B.S. and M.S. degrees in electrical engineering from University of Electronic Science and Technology of China in 1995 and 1998, respectively, and Ph.D. degree in electrical engineering from New Jersey Institute of Technology, Newark, NJ, USA in 2006. She joined Northern Illinois University in 2007 and is currently an Associate Professor of electrical engineering and the Director of the Digital Signal Processing Laboratory. Her current research interests include digital signal processing, real-time signal processing, wireless communication and networking. She has over 70 publications including 27 journal papers and one book chapter. She has three patents awarded. She served as a program committee member for the 2014 and 2010 IEEE International Conference on Networking Sensing and Control (ICNSC), and the 2009 IEEE International Conference on Service Operations and Logistics, and Informatics (SOLIS). She is the TPC members for ICC 2018, ICICT 2018, ICNSC 2009, 2010, 2012, IEEE International Conference on Mechatronics and Automation (ICMA) 2010, IEEE International Communication Conference (ICC) 2008, IEEE Globe Communication Conference (Globacom) 2009 and IEEE Wireless Communication and Networking Conference (WCNC) 2009. She is the recipient of the Best Conference Paper Finalist of IEEE Conference on Networking, Sensing and Control 2013, Best Conference Paper Finalist of IEEE Conference on Automation Science and Engineering (CASE) 2008, the Student Poster Award of Sarnoff 2005, ICNSC 2005 Best Student Paper Award, IEEE Wireless and Optical Communications Conference (WOCC) 2005 Student Poster Award and the student presentation Award of Chinese Institute of Engineer (CIE). She is the Senior Member of IEEE. She is a reviewer of *IEEE Transaction on Signal Processing*, *IEEE Transaction on Acoustic, Speech, and Language Processing*, *IEEE Transactions on Industry Electronics*, *IEEE Communication Magazine*, *IEEE Transactions on Wireless Communication* and *IEEE Communication Letters*.



Karunakar Reddy Pottim received the M.S. degree from Northern Illinois University in 2011. Karunakar Reddy Pottim's major research areas focus on active noise control and digital signal processing. He works as a program manager at Amazon since 2017.



Sen M. Kuo (M'84−SM'04) received the B.S. degree from the National Taiwan Normal University, Taipei, China, in 1976 and the M.S. and Ph.D. degrees from the University of New Mexico, Albuquerque, in 1983 and 1985, respectively.

He is currently Honorary Chair Professor in the Department of Electrical Engineering, Chung Yuan Christian University. He was a Professor in the Department of Electrical Engineering, Northern Illinois University, DeKalb, and served as the department chair from 2002 to 2008. In 1993, he was with Texas Instruments, Houston, TX, and with Chung-Ang University in 2008. He is the leading author of three books: Active Noise Control Systems (Wiley, 1996), Real-Time Digital Signal Processing (Wiley, 2001, 2007, 2013), and Digital Signal Processors (Prentice-Hall, 2005), and co-author of three books: Embedded Signal Processing with the Micro Signal Architecture (Wiley, 2007), Subband Adaptive Filtering (Wiley, 2009), and Secondary-Path Modeling Techniques for Active Noise Control Systems (Springer). He holds eight U.S. patents. His research focuses on active noise and vibration control, real-time DSP applications, digital audio applications, and biomedical signal processing.

Dr. Kuo received the IEEE 1993 Chester Sall Award for the first-place transactions (Consumer Electronics) paper award, and the faculty-of-year award in 2001 for accomplishments in research and scholarly areas.

Authorized licensed use limited to: Utah State University. Downloaded on February 03,2026 at 03:43:34 UTC from IEEE Xplore. Restrictions apply.